EduPath BD

Production Release Candidate
Feature, Technology and Manual Acceptance Guide

Version 1.0.0-rc.1
Prepared 16 September 2026
Purpose Final navigation, comparison, defect capture and deployment readiness

This manual describes what every implemented feature should do, how to test it, which technology powers it, and which production evidence still depends on external services or hardware.

1. Release verdict

Code-level result: production-hardened release candidate. Both uploaded v0.9.1 archives were byte-for-byte identical after excluding packaging paths. The current RC adds strict production configuration, separate database migrations, readiness checks, graceful shutdown, least-privilege containers, same-origin web API routing, native version alignment and bundle splitting.

Area	Verified result	What it does not prove
Baseline comparison	288 source files identical; no missing or changed application source between the two uploaded archives.	It does not prove the user laptop folder has no additional local edits.
Automated QA	Typecheck, 20 tests, TensorFlow model verification, full API contract verification, release verification, production verification and build pass.	Live Firebase, Supabase, Gemini and PostgreSQL accounts require real credentials.
Frontend build	4,102 modules; explicit chunks; largest minified chunk reduced from about 1.08 MB to 452 KB.	Real network speed and low-end device performance still require device testing.
Browser E2E	The uploaded baseline had user-provided 4/4 Playwright pass evidence.	The review container could not download a Chromium binary; rerun E2E on the laptop for the final RC.
Security audit	No critical advisory at the configured gate.	Current upstream Prisma CLI and Google Storage dependency trees still report high/moderate advisories; track upgrades.

Important: use the term **release candidate** until every external gate in Section 10 has dated evidence. Production-quality code is not the same as a publicly operated production service.

2. How to use this manual

- Start the API and web application, keep the browser Console and Network tabs open, and test one row at a time.
- Mark Pass only when the visible result, saved state, API response and reload behavior all match the expected behavior.
- For a failure, record route/section, role, exact steps, expected result, actual result, screenshot, console message and failed request.
- Repeat critical flows in English and Bangla, desktop and 360-435 px mobile widths, and after a full reload.
- Use two browser sessions for real-time and account-isolation checks; use a physical phone for notifications, safe areas and native sharing.

Recommended test order: environment and health -> public catalog -> student login -> student workspace -> community/media -> admin workflows -> offline/native -> production evidence.

3. Roles and system boundaries

Actor	Expected access	Must be denied
Guest	Public catalog, pathways, circulars, maps, verified media, PWA shell and diagnostics.	Shortlist, applications, uploads, comments, AI history and admin data.
Student	Profile, eligibility, prediction, shortlist, comparison, applications, checklist, reminders, scholarships, comments, votes, ratings, reports and AI.	Admin review, moderation, bulk import, analytics and publication actions.
Moderator	Moderation queue and report resolution plus permitted content-review actions.	Owner-only or admin-only operations not assigned by the API policy.
Admin	Content drafts, verification, media, CSV import, analytics, exports, moderation and dashboard statistics.	No bypass of validation, source URLs, audit records or verification state.
External services	Firebase identity, PostgreSQL persistence, Supabase media, Gemini responses, map/video providers.	Secrets must never be exposed to browser bundles or logs.

Architecture at a glance

Layer	Main components	Responsibility
Client	React, TypeScript, Vite, Tailwind, Framer Motion	Responsive UI, accessibility, forms, local state and lazy feature loading.
Cross-platform	Capacitor Android/iOS, PWA manifest and service worker	Native packaging, notifications, safe areas, installability and offline shell.
API	Express 5, Zod, REST, Apollo GraphQL	Validation, authorization, business rules, reporting and external-service adapters.
Real time	SSE and Socket.IO	Discussion, vote and rating events across connected clients.
Data	PostgreSQL and Prisma	Users, catalog, rules, applications, media, moderation, analytics and audit history.
Intelligence	TensorFlow.js and Gemini adapter	Offline fit scoring plus source-aware explanation and summarization.
Delivery	Docker, Compose, GitHub Actions, Playwright	Repeatable build, migrations, QA, E2E and deployment evidence.

4. Complete feature behavior and manual checks

The tables below are the main comparison reference. A feature passes only when all four columns agree with what you observe.

4.1 Product shell, localization and PWA

Feature	Expected behavior	Manual check	Red flag
Responsive shell and navigation	Desktop header, mobile menu, bottom navigation and footer reach every major section. No duplicate IDs, dead links or horizontal overflow.	Test 360, 411, 768 and desktop widths; open/close the mobile menu; use every section link; press browser Back.	Hidden auth/menu control, clipped content, wrong active state or inaccessible icon.
Bangla/English	Changing language updates visible labels and persists after reload/restart.	Switch twice, reload, sign in, open modals and inspect errors in both languages.	Mixed-language labels, lost preference or unlocalized validation.
Accessibility	Skip link, one page-level H1, named controls, visible focus, reduced-motion behavior and safe touch targets.	Keyboard-only navigation; screen-reader names; browser audit; enable reduced motion.	Focus trap, unnamed icon button, target below 40-44 px or motion that ignores preference.
PWA and offline catalog	Installable shell; previously fetched public catalogs can be shown offline; private responses are never cached.	Visit catalog online, disconnect, reload, inspect Application/Cache Storage and sign-in state.	Blank offline screen, stale private data or misleading live status.
Device diagnostics	Reports runtime, API address, readiness, storage, notification, PWA, safe area and viewport without exposing secrets.	Run on laptop and phone; deny/allow notification; stop API and rerun.	localhost used inside native app, false green status or credential leakage.

4.2 Authentication and sessions

Feature	Expected behavior	Manual check	Red flag
Username availability and reservation	Debounced availability uses the same store as registration; successful registration reserves the name; collision rolls back safely.	Try valid, invalid and existing names; submit the same name simultaneously in two sessions.	Available name later silently stolen or Firebase account left after failed reservation.
Email verification	Firebase registration requires verified email before trusted app session behavior.	Register with test Firebase project, follow verification link, retry sign-in.	Unverified identity receives protected session.
Phone OTP	Real Firebase Phone Auth in configured production; fictional-number mode only when explicitly enabled for demo.	Test fictional number and production project separately; verify reCAPTCHA and error states.	Demo bypass enabled in production or UI claims SMS when none was sent.
JWT and roles	Short-lived issuer/audience-checked token; student/moderator/admin authorization enforced server-side.	Tamper token; use wrong role on admin API; allow session to expire; reload.	UI-only protection, stale session loops or protected data returned.
Sign out	Clears saved session, updates header/mobile navigation and blocks protected reloads.	Sign in, sign out on mobile and desktop, reload, inspect local storage.	Old token remains usable or workspace stays mounted.

4.3 Discovery, eligibility and prediction

Feature	Expected behavior	Manual check	Red flag
Institution/program discovery	Verified records only; search, type/group/district filters and server pagination combine correctly.	Use each filter alone and together; navigate pages; open official source.	Draft records leak, count mismatch or filters reset unexpectedly.
Circular deadlines	Date range returns sourced verified circulars with correct ordering, fee and deadline labels.	Set valid/invalid ranges; test boundaries and empty result.	Wrong timezone day, inverted range accepted or source missing.
Eligibility engine	Server evaluates current admission-rule records by level, group and GPA; gives reasons and sourced matching programs.	Test eligible/ineligible Science, Humanities, Business and SSC technical profiles.	Client-only formula, stale result, prediction presented as eligibility.
TensorFlow pathway fit	Bundled trained 5-8-1 model runs on-device/offline and ranks stronger compatible profiles higher.	Load once, disconnect, compare strong/weak profiles and repeat result.	Network dependency, random score or claim of admission guarantee.
Why recommended	Recommendation shows reasons, matched profile factors and appropriate disclaimer.	Open several pathway results and compare reason text to inputs.	Only a percentage with no explanation or misleading certainty.

4.4 Student planning and persistence

Feature	Expected behavior	Manual check	Red flag
Three-way comparison	Up to three institutions persist locally/account context and render meaningful comparable fields.	Add 1-3, reject fourth, remove, reload and compare mobile layout.	Duplicate cards, fourth item accepted or table overflow.
Shortlist	Authenticated add/remove persists in application store and reloads correctly.	Save from catalog, verify workspace, remove, reload and check second session.	Optimistic UI disagrees with server.
Application tracker	Creates one tracked program with status flow and four-document checklist; updates persist.	Start application, toggle documents, advance status, reload.	Duplicate workflow, invalid status jump or lost checklist.
Deadline reminders	In-app reminders persist; native local notifications can schedule/cancel with honest permission state.	Create/cancel reminder; deny then grant permission; restart phone.	Permission denied shown as success or duplicate notifications.
Scholarship matching	Filters verified opportunities by level/group/GPA; saved items and source links work.	Test high/low GPA and SSC vocational; open source; save and reload.	Ineligible grant shown as match or unverifiable opportunity.
Parent progress PDF	Generates a readable student progress report and uses native share where available.	Download on desktop, share on phone, inspect dates/status/source labels.	Blank PDF, clipped rows, private token in report or unsupported share crash.

4.5 Community, rating and safety

Feature	Expected behavior	Manual check	Red flag
Discussion and replies	Authenticated comments/replies persist in own database and appear through real-time channel.	Use two sessions; post/reply; refresh both; test institution with no discussion.	False 404, duplicate message or only local UI update.
Like/dislike	One vote per account/comment; switching/removing vote recalculates server totals and broadcasts.	Vote from two accounts; switch type; reload.	Count inflation, negative count or account can vote repeatedly.
Institution rating	One 1-5 score per account; average, count and distribution update live.	Rate in two sessions; change score; inspect distribution sum.	Average computed client-side or duplicate account rating.
Report and moderation	Student report creates OPEN item; moderator/admin resolves with audit information.	Report comment, open admin queue, resolve, reload and export.	Reporter sees admin data, report disappears or no decision trail.
Rate limits and validation	Auth, writes, AI and uploads return 429 after configured limits; malformed input returns safe 4xx.	Use scripted repeated requests and invalid payloads.	Server crash, stack trace/secret returned or unbounded memory growth.

4.6 Media, maps and uploads

Feature	Expected behavior	Manual check	Red flag
Campus video precedence	Verified institution YouTube -> verified global video -> bundled EduPath admission video.	Publish resources in order, change verification state and inspect each fallback.	Unrelated hardcoded video, unverified media visible or insecure embed.
Language-aware audio	Institution language -> neutral institution -> global language -> neutral global -> configured local -> speech synthesis.	Switch app language and institution; remove each source layer in turn.	Generic pronunciation clip, 0:00 with no fallback or wrong language.
Campus map/location	Leaflet/OSM works key-free; campus marker selection and nearest listed campus use permission safely; Google embed is optional.	Allow/deny geolocation; change campus; test missing Google key.	Blank map, permission loop or Google key required for base map.
Cloud upload	Auth required; PDF/JPG/PNG/audio type, size, base64 and binary signature validated before Supabase; demo preview labelled.	Upload valid files, spoof extension/MIME, oversize, malformed base64.	Generic error, executable accepted or service-role key exposed.
Admin media review	New institution/global media is pending; only verified records are public; URLs normalized and localized.	Create YouTube/audio, inspect pending state, verify/reject and test public API.	Pending item visible or HTTP/non-direct audio accepted.

4.7 AI, administration and reporting

Feature	Expected behavior	Manual check	Red flag
Edu AI	Authenticated, source-aware answer; Gemini mode when key works; deterministic cited fallback when it does not.	Ask supported/unsupported questions with and without key; open sources.	Uncited claim, fake live-AI label or core app failure without key.
Circular summarizer	Summary remains tied to the selected verified circular and exact source ID.	Summarize two circulars and compare dates/fees/source.	Cross-circular facts or invented deadline.

Feature	Expected behavior	Manual check	Red flag
Admin content lifecycle	Create/edit draft -> review -> verify/publish; outdated parents hide child content; audit action stored.	Create institution/program/circular, validate bad dates/tuition, publish in order.	Draft leakage, child visible before parent or edit without audit.
Bulk CSV import	Quoted CSV parsed safely; valid rows become drafts; invalid rows reported without corrupting good data.	Import quoted commas, duplicate slug, bad type and mixed-validity rows.	Everything silently accepted/rejected or rows auto-published.
Analytics and exports	Protected aggregate statistics and CSV/JSON datasets reflect app activity.	Generate actions, compare dashboard, download files as admin and student.	PII leak, student access or incorrect aggregation.
Crystal Reports bridge	XSD, sample/export endpoints, PostgreSQL views and C# viewer source are available.	Open viewer on Windows and bind final .rpt to exported schema.	Claim final Crystal layout without a tested .rpt.

5. Teacher requirements: 1-24 acceptance map

#	Requirement	Pass evidence
1	PDF ERD	Open four-page vector A3 ERD; keys and relationship matrix remain readable when zoomed.
2	Own-database real-time comments	Prove PostgreSQL persistence with two clients, reload and live event.
3	Live like/dislike	One vote/account and live server totals.
4	Unique username	Debounced store lookup plus atomic registration reservation.
5	Email and phone verification	Firebase verified identities; demo number clearly labelled.
6	Textboxes/dropdowns	Student and admin forms validate and preserve accessible labels.
7	Real-time rating	One score/account; live average/count/distribution.
8	Pagination	Server metadata and mobile previous/next operate with filters.
9	External audio/video	Privacy YouTube, reviewed audio and deterministic fallbacks.
10	Maps/location	Leaflet/OSM, markers, geolocation, optional Google adapter.
11	Session/JWT	Issuer/audience token validation and role guards.
12	On-device prediction	Offline trained TensorFlow.js inference with disclaimer.
13	Cloud upload/display	Validated Supabase upload; demo preview never mislabelled production.
14	Datepicker	React Datepicker ranges and date-order validation.
15	Animation	Framer Motion transitions plus reduced-motion support.
16	Styling	Tailwind responsive system and safe mobile touch targets.
17	Git/Vite workflow	Reproducible install, build, QA and hosted history.
18	GraphQL	Apollo institution query returns filtered data.
19	REST/SOAP requirement	Express REST coverage demonstrated; explain modern REST choice if SOAP is not used.
20	Crystal Report	Viewer/schema/export exist; final .rpt needs Windows proof.
21	Modern frontend	Shared React/TypeScript/Capacitor Android, iOS and PWA source.
22	Docker/CI/CD	Migration job, secure API runtime, readiness and green hosted workflow.
23	Generative AI	Gemini-labelled live response with sources; fallback honesty.
24	Additional features	Localization, offline, scholarship, reports, import, moderation, analytics and more.

6. Tools and technologies - what each one builds

Technology	Primary role	Features built
React	Component UI	Shell, cards, dialogs, student/admin workspaces and stateful interaction.
TypeScript	Static safety	Shared data contracts, safer refactors and compile-time errors.
Vite / Rolldown	Build and dev server	Fast development, production assets, preview proxy and explicit vendor chunks.
Tailwind CSS	Design system	Responsive spacing, colors, typography, touch targets and dark product shell.
Framer Motion	Animation	Page/card/result transitions with reduced-motion awareness.
Capacitor	Native bridge	Android/iOS packaging and access to native capabilities from shared web code.
PWA + service worker	Install/offline	Installable web shell, public catalog caching and connectivity behavior.
Express 5	REST API	Auth, catalog, eligibility, uploads, AI, admin and reporting routes.
Zod	Request validation	Rejects invalid filters, forms, dates and payloads before domain logic.
Apollo GraphQL	GraphQL API	Modern typed institution query foundation.
Socket.IO + SSE	Real time	Discussion/rating event distribution and simple stream verification.
PostgreSQL	Primary database	Durable relational data for users, catalog, workflows, community and audit.
Prisma	Data access/migrations	Schema, typed client, committed migrations, seed and production deploy.
Firebase Auth/Admin	Identity	Email/phone verification and server-side Firebase token validation.
Supabase Storage	Media storage	Cloud file persistence behind server validation and role enforcement.
Gemini adapter	Generative AI	Source-aware answers and circular summaries when a key is configured.
TensorFlow.js	On-device ML	Offline pathway-fit model with committed reproducible artifact.
Leaflet / React-Leaflet	Maps	OpenStreetMap campus maps, markers and device location without a paid key.
React Datepicker	Date input	Accessible student ranges and admin admission dates.
Capacitor Local Notifications	Native reminders	Schedule and cancel deadline alerts on supported devices.
jsPDF + html2canvas	PDF reporting	Parent/student progress report generation and sharing.
Docker / Compose	Runtime packaging	Postgres, migration stage and least-privilege API service graph.
GitHub Actions	CI	Clean install, QA, browser E2E, audit and retained failure evidence.
Node test runner	Unit/integration tests	JWT, config, registration, eligibility, CSV and upload checks.
Playwright	Browser acceptance	Desktop/mobile navigation, console/network audit and demo flow.
Helmet / CORS / JWT / throttling	Security controls	Headers, exact origins, signed sessions and abuse reduction.
Crystal Reports + C# viewer	Legacy reporting requirement	Consumes protected exports using XSD/views and Windows viewer source.

7. High-value end-to-end test journeys

Journey	Steps and expected outcome
A. New student decision journey	Guest -> switch language -> browse programs -> sign in -> complete profile -> eligibility -> pathway fit -> inspect reasons -> compare three -> shortlist -> scholarship match -> track application -> document checklist -> reminder -> parent PDF.
B. Trust and source journey	Open institution/circular/scholarship source -> inspect verified badge and update date -> compare UI result with official page -> report a questionable comment -> verify moderator resolution.
C. Real-time journey	Open student A and student B -> join same institution -> comment -> reply -> like/dislike -> rating -> verify live totals in both -> reload -> verify persistence.
D. Media journey	Select institution -> verify map -> institution video/audio -> remove dedicated resource -> global fallback -> bundled video/device speech -> publish new reviewed media as admin.
E. Admin publishing journey	Admin login -> create institution draft -> create child program -> confirm public hidden -> verify parent -> verify child -> edit source/date -> inspect audit -> CSV import -> analytics/export.
F. Resilience journey	Expire token -> protected calls clear session once -> stop API -> cached public catalog/offline shell -> restore API -> readiness turns green -> no duplicate captured error.
G. Native phone journey	Install debug build with LAN API -> safe area -> sign in/out -> notifications -> geolocation -> upload -> share PDF -> rotate -> back button -> offline/reconnect.

8. Better defect-catching workflow

Stage	Action
1. Automated gate	Run `npm run qa`, `npm run test:e2e` and `npm run audit:critical`. Save terminal output.
2. Browser diagnostics	Open the in-app QA panel, clear captured errors, run checks, navigate one journey, export JSON.
3. DevTools	Keep Console and Network visible. Filter failed requests, preserve log and disable cache only while diagnosing.
4. Two-session test	Use normal + incognito browsers for identity isolation and real-time features.
5. Device matrix	At minimum: desktop Chrome, mobile emulation 360 px, physical Android portrait/landscape.
6. Evidence packet	For each failure attach screenshot/video, QA JSON, Playwright trace if available, route, account role, API mode and exact timestamp.
7. Retest discipline	Fix one issue, add or update an automated regression check, rerun the smallest test, then the full QA gate.

9. Manual acceptance checklist

- 1. Fresh install/start shows no blank screen, crash or repeated console error. **Pass / Fail / N/A:** _____
- 2. Health reports alive; readiness reports ready; API version is 1.0.0-rc.1. **Pass / Fail / N/A:** _____
- 3. Every desktop/mobile navigation target exists after lazy content settles. **Pass / Fail / N/A:** _____
- 4. English/Bangla preference persists after reload and native restart. **Pass / Fail / N/A:** _____
- 5. Guest, student, moderator and admin boundaries match Section 3. **Pass / Fail / N/A:** _____
- 6. Username, email and phone states are honest and collision-safe. **Pass / Fail / N/A:** _____
- 7. Catalog totals, filters, pagination and sources match API results. **Pass / Fail / N/A:** _____
- 8. Eligibility rules and ML prediction are visibly different concepts. **Pass / Fail / N/A:** _____
- 9. Comparison, shortlist, applications and documents persist correctly. **Pass / Fail / N/A:** _____
- 10. Scholarships, deadlines, datepicker and reminders behave at boundaries. **Pass / Fail / N/A:** _____
- 11. Two clients prove comments, replies, votes and ratings persist/live-update. **Pass / Fail / N/A:** _____
- 12. Reports reach moderation and cannot expose admin data to students. **Pass / Fail / N/A:** _____
- 13. Map, video and audio follow documented fallback precedence. **Pass / Fail / N/A:** _____
- 14. Uploads reject spoofed, malformed, oversized and unsupported files. **Pass / Fail / N/A:** _____
- 15. AI answers cite sources and label Gemini/fallback mode correctly. **Pass / Fail / N/A:** _____
- 16. Admin draft/review/publish hierarchy prevents unverified data leakage. **Pass / Fail / N/A:** _____
- 17. CSV import reports bad rows and never auto-publishes drafts. **Pass / Fail / N/A:** _____
- 18. PDF/CSV/JSON reports download and contain expected fields without secrets. **Pass / Fail / N/A:** _____
- 19. Offline shell/public cache works; private responses are absent from cache. **Pass / Fail / N/A:** _____
- 20. Phone safe area, rotation, Back, keyboard, geolocation, share and notification work. **Pass / Fail / N/A:** _____
- 21. No horizontal overflow at 360 px and all primary controls meet touch size. **Pass / Fail / N/A:** _____
- 22. Production config refuses missing/placeholder secrets and insecure origins. **Pass / Fail / N/A:** _____
- 23. Migration job completes before API; production startup never auto-seeds. **Pass / Fail / N/A:** _____
- 24. Hosted CI, backups, alerts, domain/TLS, Android signing and external services have evidence. **Pass / Fail / N/A:** _____

10. External production gates

Gate	Required launch evidence	Owner/date
PostgreSQL	Run migration and full student/admin flows with DEMO_MODE=false; capture persistence after restart.	_____
Official data	Replace demo circulars and small catalog with current sourced records; assign reviewer/update dates.	_____
Firebase	Production project, authorized domains, verified email and real/test phone evidence.	_____
Supabase	Private bucket policies, upload/display/delete behavior and secret isolation.	_____
Gemini	Live labelled response, quota behavior, safety and fallback test.	_____
HTTPS/domain	Public TLS, reverse proxy for /api, /graphql and /socket.io, exact WEB_ORIGINS.	_____
Android release	Release keystore backup, signed AAB, release-mode device test and Play Console policy assets.	_____
iOS release	macOS/Xcode signing, archive and physical-device acceptance.	_____
CI/CD	Green hosted workflow on final commit and retained Playwright artifacts.	_____
Operations	Encrypted backup, successful restore drill, availability/error/latency alerts and rollback image.	_____
Legal/privacy	Privacy policy, terms, retention, account deletion and consent review.	_____
Crystal Reports	Final .rpt authored and opened on Windows with actual exported dataset.	_____
Dependency advisories	Track and retest compatible Prisma/Google dependency upgrades; keep audit output visible.	_____

11. Bug report template

ID / severity / date: _____

Build version and commit/archive hash: _____

Device, OS, browser and viewport: _____

Role and account: _____

API mode and URL: _____

Section/route: _____

Preconditions: _____

Exact reproduction steps: _____

Expected result: _____

Actual result: _____

Console error: _____

Failed network request/status: _____

Screenshot/video/QA JSON/trace: _____

Frequency: _____

Workaround: _____

Regression test added: _____

Retest result: _____

12. Final sign-off

Release label: **EduPath BD v1.0.0-rc.1**

Decision	Meaning
Accepted for classroom presentation	Core QA is green; presentation dependencies are available; open external production gates are disclosed.
Accepted for limited pilot	Real database/auth/storage/domain, monitoring and backup evidence is complete for a controlled user group.
Accepted for public production	Every Section 10 gate has dated evidence, responsible owner and rollback plan.
Rejected	Any critical security, data integrity, auth bypass, crash, blank screen or misleading source/AI result remains.

Tester: _____ Date: _____

Decision: _____ Critical issues open: _____

Signature: _____ Evidence folder: _____

End of manual

Keep this PDF with the exact source archive and SHA-256 checksum used for testing.